

Introduction to Numerical Methods 11

1

Boundary value problems

Problem to solve: $\frac{dx}{dt} = f(x, t)$, $x(t_0) = x_0$ $x(t) = ?$

Discrete approximation

$$dt \rightarrow \Delta t, \quad dx \rightarrow \Delta x$$

$$t_n = t_0 + n \Delta t$$

Euler method

$$x_{n+1} = x_n + f(x_n, t_n) \Delta t$$

Stability of the Euler method

eg. $f(x, t) = -\lambda x$

$$x_{n+1} = (1 - \lambda \Delta t) x_n$$

converges if $|1 - \lambda \Delta t| < 1$

Any dimensions: $\frac{dx}{dt} = f(x, t)$

$\hookrightarrow$ eigen system: the most important is the

largest eigen value: λ

Euler method converges if $|1 + \lambda \Delta t| < 1$

Implicit Euler

$$x_{n+1} = x_n + f(x_{n+1}, t_{n+1}) \Delta t$$

we need to solve it for x_{n+1}

in case of $f(x, t) = -\lambda x$

Stability of the Implicit Euler for $f(x) = -x^2$

$$k_{eff} = \frac{h}{1 - 2\Delta t}$$

which is stable for $h > 0 \Rightarrow \Delta t$ is stable

System is stable if
Error $O(h)$

Higher order methods

(Least)

Runge-Kutta method

$$y_{n+1} = y_n + \frac{h}{6} (k_1 + 4k_2 + k_3)$$

Runge-Kutta

$$k_1 = f(t_n, y_n)$$

$$k_2 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2} k_1)$$

$$k_3 = f(t_n + \frac{h}{2}, y_n + h k_2)$$

$$k_4 = f(t_n + h, y_n + h k_3)$$

Slope at the beginning

Slope at midpoint

Slope at other midpoint

Slope at the end

$$Error = O(h^4)$$

But is implicit Euler

more precise - always stable

$O(h^2)$ - less calculations

and this

order, more

with stability as further evaluation

in many applications (H.Y.)

Second order differential equation

(Newton's equation of motion)

$$\ddot{x} = \frac{F}{m} = f(x, t), \quad x(t=0) = x_0, \quad v(t=0) = v_0$$

Euler method

$$v_{n+1} = v_n + f(x_n) \Delta t$$

$$x_{n+1} = x_n + v_n \Delta t$$

Implicit Euler method

for position use new velocity

$$v_{n+1} = v_n + f(x_n) \Delta t$$

$$x_{n+1} = x_n + \boxed{v_{n+1}} \Delta t$$

Symplectic integrator: conserves

the structure of the equations,

keeps energy (including some virtual terms) constant

The Euler method is not symplectic

The implicit Euler is a symplectic integrator

Higher order methods

Runge-Kutta works but is not a symplectic integrator.

General standard: Velocity Verlet

$$x(t + \Delta t) = x(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2$$

$$v(t + \Delta t) = v(t) + \frac{a(t) + a(t + \Delta t)}{2} \Delta t$$

Algorithm:

1. $x(t + \Delta t) = x(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2$
2. get $a(t + \Delta t)$ using the interactions at $x(t + \Delta t)$
3. $v(t + \Delta t) = v(t) + \frac{1}{2}(a(t) + a(t + \Delta t))\Delta t$

Also a symplectic integrator